QUBO Formulation: Exact Cover by 3-Sets Problem

Exact Cover by 3-Sets Problem

Problem Statement. Let U be a finite universe of N elements, and let $\mathcal{S} = \{S_0, S_1, \ldots, S_{M-1}\}$ be a collection of M subsets of U, where each $S_j \subseteq U$ satisfies $|S_j| = 3$. The Exact Cover by 3-Sets Problem asks whether there exists a subcollection $\mathcal{C} \subseteq \mathcal{S}$ such that every element $i \in U$ belongs to exactly one set in $\mathcal{C}$. If such a subcollection exists, it is called an exact cover. The optimization task is to select subsets to minimize the total coverage violations across all universe elements, where a violation count of zero indicates a valid exact cover.

Decision Variables. For each set S_j with index $j \in \{0, \ldots, M-1\}$, introduce a binary decision variable $x_j \in \{0, 1\}$. The variable $x_j = 1$ indicates that set S_j is selected for the cover, and $x_j = 0$ indicates it is not. The decision vector is denoted $\mathbf{x} = (x_0, x_1, \ldots, x_{M-1}) \in \{0, 1\}^M$.

QUBO Derivation. Step 1 — Original Objective. The objective is to minimize the sum of squared deviations from the exact cover condition across all universe elements. We define the Hamiltonian $H(\mathbf{x})$ directly as this sum:

$$H(\mathbf{x}) = \sum_{i \in U} \left(\sum_{j : i \in S_j} x_j - 1 \right)^2 . \tag{1}$$

Step 2 — Constraint Form. For each universe element $i \in U$, the exact cover requirement is enforced by the linear equality:

$$\sum_{j : i \in S_j} x_j = 1, \quad \forall i \in U. \tag{2}$$

Let $\mathcal{N}_i = \{j \in \{0, \ldots, M-1\} : i \in S_j\}$ denote the set of indices of subsets containing element i. The constraint is equivalently written as $\sum_{j \in \mathcal{N}_i} x_j = 1$.

Step 3 — Penalty Expansion. We expand the squared penalty for each element $i \in U$

using the idempotent property $x_j^2 = x_j$ for binary variables:

$$\begin{aligned}\left(\sum_{j\in\mathcal{N}_i} x_j - 1\right)^2 &= \left(\sum_{j\in\mathcal{N}_i} x_j\right)^2 - 2\sum_{j\in\mathcal{N}_i} x_j + 1 \\ &= \sum_{j\in\mathcal{N}_i} x_j^2 + \sum_{\substack{j,k\in\mathcal{N}_i \\ j\neq k}} x_j x_k - 2\sum_{j\in\mathcal{N}_i} x_j + 1 \\ &= \sum_{j\in\mathcal{N}_i} x_j + \sum_{\substack{j,k\in\mathcal{N}_i \\ j\neq k}} x_j x_k - 2\sum_{j\in\mathcal{N}_i} x_j + 1 \\ &= 1 - \sum_{j\in\mathcal{N}_i} x_j + \sum_{\substack{j,k\in\mathcal{N}_i \\ j\neq k}} x_j x_k. \end{aligned} \tag{3}$$

Step 4 — Combined Hamiltonian. Summing the expanded penalties over all $i \in U$ yields:

$$\begin{aligned} H(\mathbf{x}) &= \sum_{i\in U}\left(1 - \sum_{j\in\mathcal{N}_i} x_j + \sum_{\substack{j,k\in\mathcal{N}_i \\ j\neq k}} x_j x_k\right) \\ &= \sum_{i\in U} 1 - \sum_{j=0}^{M-1}\left(\sum_{i\in U}\mathbb{I}(i\in S_j)\right)x_j + \sum_{j=0}^{M-1}\sum_{\substack{k=0 \\ k\neq j}}^{M-1}\left(\sum_{i\in U}\mathbb{I}(i\in S_j\cap S_k)\right)x_j x_k, \end{aligned} \tag{4}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Since each set has exactly three elements, $\sum_{i\in U}\mathbb{I}(i \in S_j) = 3$. The quadratic term sums over ordered pairs (j,k) with $j \neq k$. Converting to the standard upper-triangular QUBO form $\sum_{j<k}$ introduces a factor of 2 for each unordered pair:

$$H(\mathbf{x}) = N - 3\sum_{j=0}^{M-1} x_j + \sum_{j=0}^{M-1}\sum_{\substack{k=0 \\ k>j}}^{M-1} 2|S_j\cap S_k| x_j x_k. \tag{5}$$

Step 5 — Q Matrix and Offset. Reading off the coefficients from the general closed form above, the QUBO matrix entries $Q_{j,k}$ and constant offset c are:

$$Q_{j,j} = -3, \quad \forall j \in \{0,\ldots,M-1\}, \tag{6}$$

$$Q_{j,k} = 2|S_j\cap S_k|, \quad \forall j < k, \tag{7}$$

$$c = N. \tag{8}$$

Penalty Weight Justification. The problem specification defines the objective function itself as the sum of squared penalties for the exact cover constraints. Consequently, no external penalty weight λ is introduced or required. The quadratic form inherently penalizes

any deviation from the constraint $\sum_{j \in \mathcal{N}_i} x_j = 1$ by exactly the square of the violation magnitude. Because the objective is to minimize this sum, any infeasible assignment incurs a strictly positive energy cost proportional to the squared violation, ensuring that feasible exact covers (energy 0) are strictly preferred over infeasible ones without the need for heuristic weight tuning.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0,1\}^M$ recovers the optimal solution because $H(\mathbf{x})$ is constructed as a sum of non-negative squared terms. For any feasible exact cover, each constraint is satisfied exactly, yielding $H(\mathbf{x}) = 0$. Since $H(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, the global minimum must be 0. Any infeasible assignment violates at least one constraint, resulting in $H(\mathbf{x}) > 0$. Thus, the ground state of H corresponds exactly to a valid exact cover if one exists, and the minimum energy value directly indicates feasibility (zero) or the total squared violation otherwise.

Illustrative Example. Consider a small instance with $N = 4$ universe elements $U = \{0,1,2,3\}$ and $M = 3$ sets $S_0 = \{0,1,2\}$, $S_1 = \{0,1,3\}$, $S_2 = \{0,2,3\}$. Each set has size 3. The pairwise intersections are $|S_0 \cap S_1| = 2$, $|S_0 \cap S_2| = 2$, and $|S_1 \cap S_2| = 2$. Instantiating the general formulas yields:

$$Q_{0,0} = Q_{1,1} = Q_{2,2} = -3, \quad Q_{0,1} = Q_{0,2} = Q_{1,2} = 4, \quad c = 4. \tag{9}$$

The concrete Hamiltonian is:

$$H(\mathbf{x}) = 4 - 3(x_0 + x_1 + x_2) + 4(x_0x_1 + x_0x_2 + x_1x_2). \tag{10}$$

Evaluating $H(\mathbf{x})$ over all $2^3 = 8$ binary assignments gives:

$\mathbf{x}$	$H(\mathbf{x})$
$(0,0,0)$	4
$(1,0,0), (0,1,0), (0,0,1)$	1
$(1,1,0), (1,0,1), (0,1,1)$	2
$(1,1,1)$	1

(11)

The minimum energy is 1, achieved by $\mathbf{x} = (1,0,0)$, $\mathbf{x} = (0,1,0)$, and $\mathbf{x} = (0,0,1)$. These correspond to selecting exactly one set, which covers three elements exactly once and leaves one element uncovered, resulting in a total squared violation of 1. No assignment yields energy 0, confirming that no exact cover exists for this instance.